Developing the Reinforcement-Learning Child Agents for Measuring Play and Learning Performance in Kindergarten Design

Jin Lee[1], Seung Wan Hong[2]
[1,2]Inha University, Incheon, Republic of Korea
[1,2]{jlee|hongsw}@inha.ac.kr

Although kindergarten design requires promoting play behaviours related to physical and social development and ensuring safety of children to support a child-oriented design, the systemic evaluation of design performance remains challenging because of the children's dynamic play behaviours. As one solution, agent-based simulations have been applied in the design field, but there are limitations in reflecting children's behavioural richness. To overcome this shortcoming, this study developed reinforcement learning (RL) child agents to compute the dynamic play behaviours associated with physical and social development. Several iterations were conducted to implement the RL agents' play behaviours, and the results were incorporated into the simulation. To validate the play behaviour model, we conducted a case analysis with authentic and unbuilt kindergarten designs and measured the quantifiable design performance in terms of physical and social play behaviours and safety. The results indicated that the RL child agents enabled a holistic analysis and the calculation of generative behavioural responses, depending on physical variations. By facilitating an unknown design affordance, the RL-powered simulation model is expected to provide data-driven evidence to support a child-oriented design.

Keywords: *Agent-based Simulation, Reinforcement-learning Agent, Children's Play Behaviour Model, Quantifiable Design Performance, Child-oriented Learning Environment*

INTRODUCTION

The design of children's learning environments has undergone significant changes and has evolved into a more child-oriented design, playing a crucial role in facilitating children's physical and social development. In particular, the child-oriented environment aims to support the heterogeneous stages and factors associated with children's development by facilitating age-specific learning through active interaction with the surrounding context and peers (Montessori, 1912). Therefore, play behaviours has been currently highlighted as an important factor which provides relevant motor skills and social facilitation (Singer et al., 2006).

However, architects face challenges in designing such environments due to the unpredictable nature of children's play behaviours. Current design methodologies, such as children's participatory designs and post-occupancy evaluation, rely on habitual and deductive extrapolation. These methods have limitations in evaluating the performance of a final product and reflecting on physical and social affordances and individual characteristics in dynamic situations (Kalay, 2004).

To overcome these challenges, agent-based simulations equipped with virtual users (VUsers) are utilised in the design field to analyse and predict the physical and social performances of space with observable behavioural responsiveness and analytics (Kalay, 2004). However, current agent-based behavioural models are primarily based on animation clips, which result in rigid behavioural features predetermined by the operator and limit their ability to reflect children's behavioural richness.

In this paper, we propose a new behavioural model using the reinforcement learning (RL) approach to compute children's divergent play behaviours. The RL agent learns how to behave on its own through a computational learning process. This approach is widely used to enhance locomotion and generate more localised responses to given settings (Peng et al., 2018). The RL child agent implemented physical and social play behaviours through the parameterisation of motor skills and social dynamics. Using this new behavioural modelling approach, this study aims to investigate the effectiveness of RL child agents in measuring the play and learning performances of a kindergarten design via overall analytics.

RELATED WORK

Behaviours in agent-based simulations

An agent-based simulation is activated by autonomous, rational computer agents called virtual users (VUsers), who compute goals, social characteristics, cognition and physical abilities of building users in a virtually constructed environment (Kalay, 2004). Theses VUsers represent the use of space and enable quantitative measurement of the design and analytics, such as circulation, walking distance, wayfinding, bottlenecks and collisions.

To capture the socio- and spatial interrelationships of the real world, the current behavior model of VUsers is based on predetermined motion data (Yan & Kalay, 2006). However, when it comes to predicting children's varying responses and evaluating their bodily experiences, manually designated motion is insufficient because of children's unpredictable behavior ranges. The rule-controlled actions often fail to adapt to an unconventional situation and represent behavioural richness in situations involving dynamic motion (Peng et al., 2018). To address such shortcomings, a new behaviour model should be explored to support the complex, autonomous children's play behaviours.

Constructing divergent behaviour responses

The RL agent is based on a machine-learning computer algorithm through which an agent gradually learns its behavioural outcome by performing a given task and receiving relevant rewards. The jointly constrained RL agent is currently considered a successful solution for the high-dimensional description of locomotion behaviour (Peng et al., 2018).

Unlike animated motion, the RL agent is a highly flexible approach that implements physics-based dynamic motor skills through task-specific control (Figure 1). It is widely adopted in robotics to improve contact-rich manipulation (Beltran-Hernandez et al., 2020). It is also used to generate dynamic movement and to control biped locomotion and physics skills in computer animation. Therefore, many studies have adopted RL agents to reproduce muscle contraction dynamics and parameterise RL actions to improve locomotion (Lee et al., 2019).

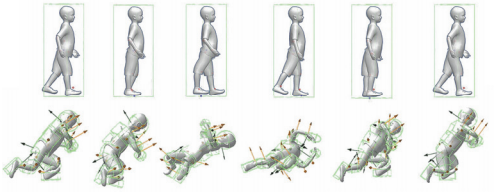


Figure 1
Comparison of behaviours: (top) animated motion, (down) RL

Nevertheless, the RL agent has rarely been applied in the context of architectural design. Since the inspiration for the design product emerges from perceiving the undiscovered affordance of the human body (Boden, 2010), the subtle behavioural

response of the RL agent is expected to have a powerful application. Therefore, this study proposes the RL agents with trained play behaviours for holistic performance measures of kindergarten design.

DEVELOPMENT OF REINFORCEMENT LEARNING CHILD AGENTS

Overview

The RL child agent is a computer agent that learns physical and social behaviours through the rewards function (Algorithm 1). This agent has 16 segmented joints that allow for subtle behavioural control with the inputs of mass, movable joint degree and angular velocity. The agent visualises diverse behavioural responses, such as falling over and implementing different mobilities based on geometric properties.

```
Algorithm 1: RewardsFunctionForTraining
a = ActionBuffer.ContinuousActions
reward = 0
function OnActionReceived():
    for each bodyPart in bodyParts:
     bodyPart.JointRotation(a[i++], a[i++], a[i++])

float CalculateRewardsForPhysicalPlay():
     if  bodyParts[foot].touchingSlopes:
        reward = 1
    else:
        reward = -1
        return reward

float CalculateRewardsForSocialPlay():
     prameters.TeamId == Team.AgentA:
     if  Collision.Ball.BodyParts[foot]
        reward = 0.2
     if  Collision.Ball.CompareTag("Goal"):
       reward = 1
      return reward
```

The key parameters of the agent included human- and environment-related factors. Physical and cognitive perspectives acted as supporting sub-parameters (Figure 2). The agent learned physical play behaviours through the interactions of (1) physical human parameters (body size, strength and speed) and (2) the physical environment (curvature, angles of surface and material friction). The social play behaviour was implemented through (1) the cognitive human parameter (visual perception) and (2) the cognitive environment (multi-agent environment).

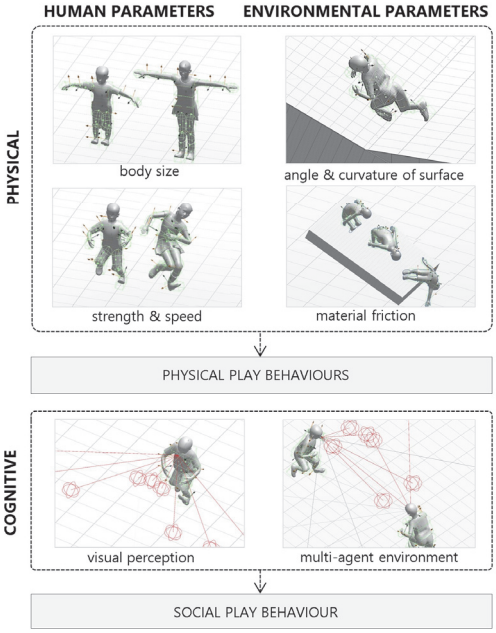


Figure 2
Parameters for training play behaviours

As an implementation tool, we utilised the commercial 3D simulation platforms *Unity 3D* and *Unity ML-Agents Toolkit* (Juliani et al., 2018). For the deep learning processes, an external *Python API* was used to implement *Tensorflow*. We coded a learning pipeline using a *C#* script and used a learning algorithm based on proximal policy optimisation.

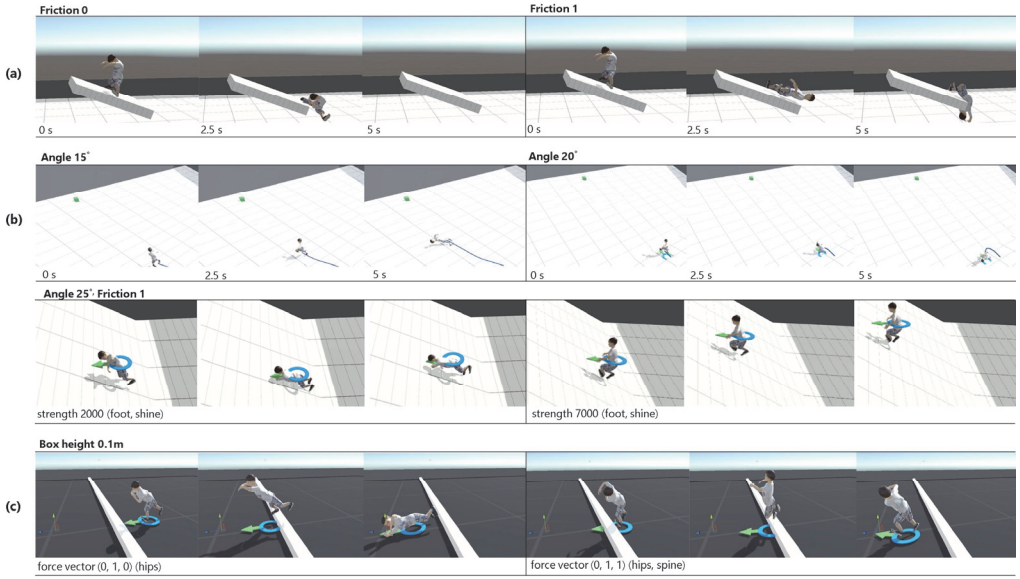


Figure 3: Physical play training

Training in physical play behaviour

Understanding the physical mechanism of human movement was an essential phase of the development to train the model for physical play behaviours. To determine the appropriate physical values and behavioural parameters, we thoroughly examined (1) the related environmental parameters, (2) strength and (3) the key joints for movement.

First, we tested the behavioural responses to the physical environment to determine the range of play behaviours. For example, we detected how the agent reacted to different levels of friction, ranging from 0 to 1. When the friction value was set to 0, the agent fell from the surface after 1.5s. In contrast, when the friction value was 1, the agent received a strong support from the surface and fell after 2.5s (Figure 3-(a)).

Next, we examined the maximum range of slope angles that the agent could climb based on the initial agent response. The results showed that the agent can cover slope angles between 15^0 uphill and -20^0 downhill. To train the agent to cover the steeper angles of the slopes, the agent was tested by applying different ranges of strength input values to the feet and shins. We set a physical environment with a slope of 25^0 and a friction value of 1. When the strength value was set to 7000, the agent successfully climbed the slope with both feet (Figure 3-(b)).

Furthermore, we conducted ergonomic tests to understand the mechanical movement and to identify the physical play behaviours that could be trained through RL. For example, we utilised an environment with a rectangular box of 0.1m height to determine the key body parts and force vectors for jumping. The agent fell over when a vertical force (0, 1, 0) was applied to the hips. When we applied a force vector (0, 1, 1) to the hips and spine, the agent was able to successfully jump over the box (Figure 3-(c)). Through these iterations, we defined key body parts, strengths and physical play ranges of the RL child agent.

Training in social play behaviour

To train in social play behaviours, an optical sensor was installed in the agents. This sensor had a vision range from -30^0 to 30^0, thereby allowing the agents to interact in adversarial situations, particularly gameplay situations. Prior to training, we conducted a test to train the agents in the game of hide-and-seek by enabling them to recognise one another (Figure 4-(a)).

As our social play of focus, we chose ball play and utilised a self-play-based RL algorithm specifically designed for the multi-agent environment in the *Unity ML-Agents Toolkit*. This algorithm assigned an ID to both the agent (self) and the other, thereby creating a multi-agent environment. To be aware of this social environment, an optical sensor was utilised in this case to detect the position of the peer, the ball and the target goal points. The agents considered not only the visual observation but also the vector spaces, including speed values of the ball along the x, y and z directions and the positions of the surrounding walls (Figure 4-(b)).

Simulating RL child agents

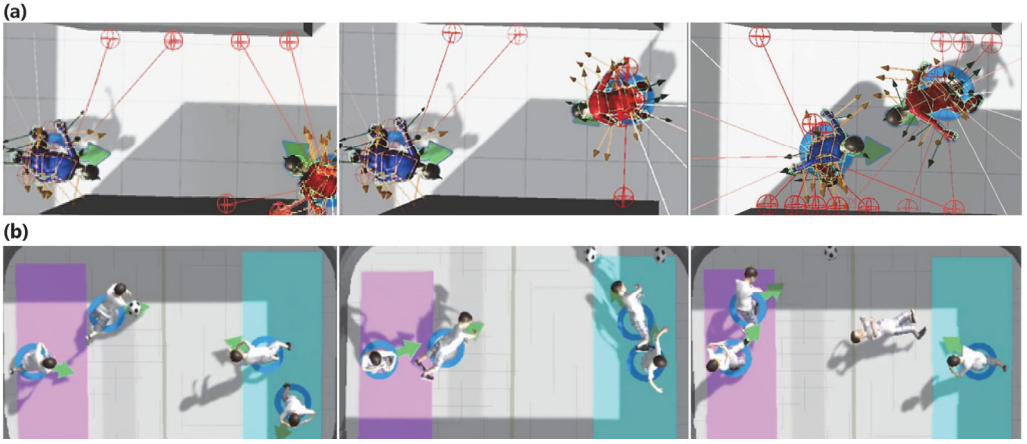
RL-based behaviour models were integrated into an agent-based simulation (Algorithm 2). The simulation was programmed to assess quantifiable learning performances regarding physical and social play behaviours and safety (Figure 5). Considering the critical ages for learning the fundamental movement skills and developmental stages (Frost et al., 2012), the simulation had two types of agents—younger (3-4 years) and older children (5-6 years) with selected behaviours. The simulation incorporated RL-based dynamic play behaviours for (1) individual physical play (e.g. walking, climbing, sliding and crawling) and (2) social play (e.g. imitation (following) for younger children and ball play for older children). Additionally, the simulation included static behaviours, such as sitting and reading, created by *Autodesk 3DS Max*.

Algorithm 2: IntegratingRLmodel
function PerceiveEnvironment():
    if BodyParts[foot].touchingSlopes = true:
        climbingStart = true
        Climbing()

function Climbing():
    SetModel(ClimbBehaviour, ClimbRLdata)
    PlayAttemptCount++

Measuring physical development: For physical play measures, the algorithm computes the frequency of physical attempts and the number of interferences (the duration of physical play). For example, the algorithm refers to exercise rates for

Figure 4
Social play training



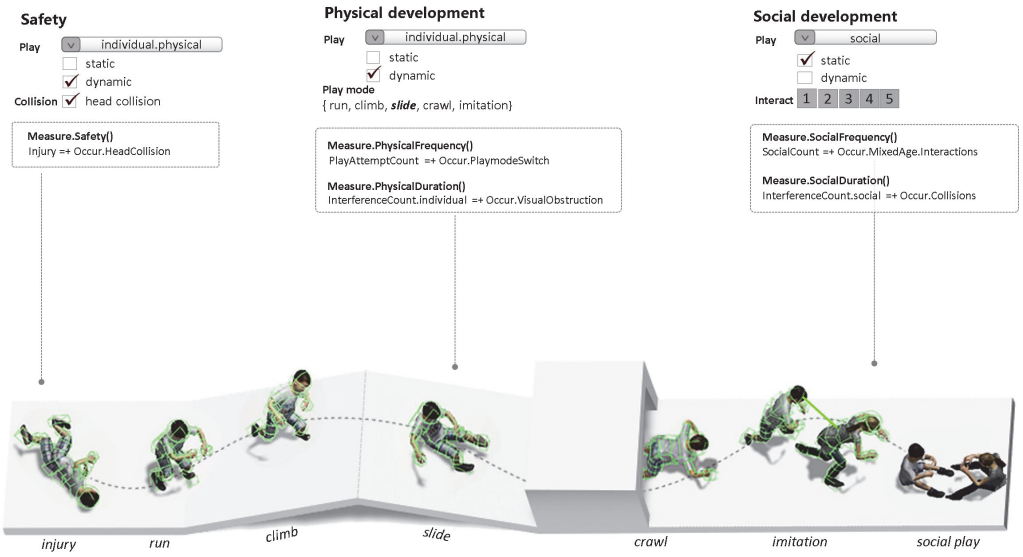


Figure 5
Mechanism of the simulation

the number of play attempts. The agents perceive the physical environments through body sensors attached to each joint and an additional visual perception of 8m. With these sensors, the agents try to climb or slide on the angular surfaces based on the friction values and demonstrate crawling behaviour in the cave-like space. Also, the system analyses concentration during playtime by calculating the interference based on the number of visual obstructions between static and dynamic play behaviours.

Measuring social development: The algorithm quantitatively measures social play with (1) the number of social play occurrences and (2) the number of interferences (the duration of social play) amongst mixed age groups and visualises the point of social play occurrence. For example, it stores the experience of social play by calculating the number of interactions between older and younger agents. Once the interaction count surpasses the input value of 5, the younger child starts following the older child and begins social playing when the younger child is within a certain distance from the older child. The system also monitors interference from other agents during social play and allows agents to start individual physical play when the number of interferences is more than three.

Measuring safety: The simulation calculates the number of injuries (head collision) and visualises the collision point. The system detects injuries when the collider attached to the agent's head hits the ground.

Case study

The simulation model offers three analytic modes. The circulation mode uses magenta and cyan colours to indicate the younger and older children's circulations, respectively. The social network mode uses color-coded circles to show changes in social relationships, with cyan for agents within 4m, blue for mixed-age social play, and red for play interference. The occurrence mode tracks the occurrence points of injury and mixed-age social plays with red and green markers, respectively.

To validate the RL child agents, Taka Tuka Land kindergarten in Germany, planned by Baupiloten,

was used as an empirical case study (Figure 6). In this case, physical and social play occurred 25 and 12 times, respectively, with the central corridor and outdoor climbing structures being the most interactive places. The RL child agents took different speeds depending on physical variations, which resulted in where the interactions occurred.

Safety analysis revealed an average of 10 head collisions which occurred mostly in the central corridor and the outdoor climbing structures due to bottlenecks and children moving from indoors to outdoors. The RL child agents enabled addressing both scenarios relevant to safety: (1) when the physical environment posed challenges and (2) when the child encountered interference from others. Through this empirical case analysis, we confirmed that this simulation provided insights into dynamic play and injury situations with children's holistic experiences and quantified performance measures with explicit evidence.

EXPERIMENTATION AND RESULTS

Experimentation

This study investigated the effects of using RL child agents to conduct a comparative lab experiment with and without simulations. The experiment involved 14 professional architects (11 males and 3 females), with an average age of 38 years. Participants were given the task of designing a kindergarten with the capability to accommodate around 60 younger and older children and a total area of 320m^2: an activity area (100 m^2), an outdoor playground (100 m^2) and younger and older children's classrooms (60 m^2 each). During the design process, participants were required to create design alternatives for a safe play environment, promoting physical and social play between younger and older children.

The experiment involved two sessions, each lasting 2h. In the first session, participants used traditional methods (sketching, diagrams and 3D modellers) to complete and modify their designs. In the second session, they utilised the simulation model to re-evaluate and develop their initial design alternatives. Quantifiable performance data of the proposed designs were collected during the experiments using simulations. Afterwards, we conducted a retrospective interview and self-assessment survey to gather insights into the participants' experiences of the design process, with and without simulation. In total, 17 kindergarten design options were generated. choose

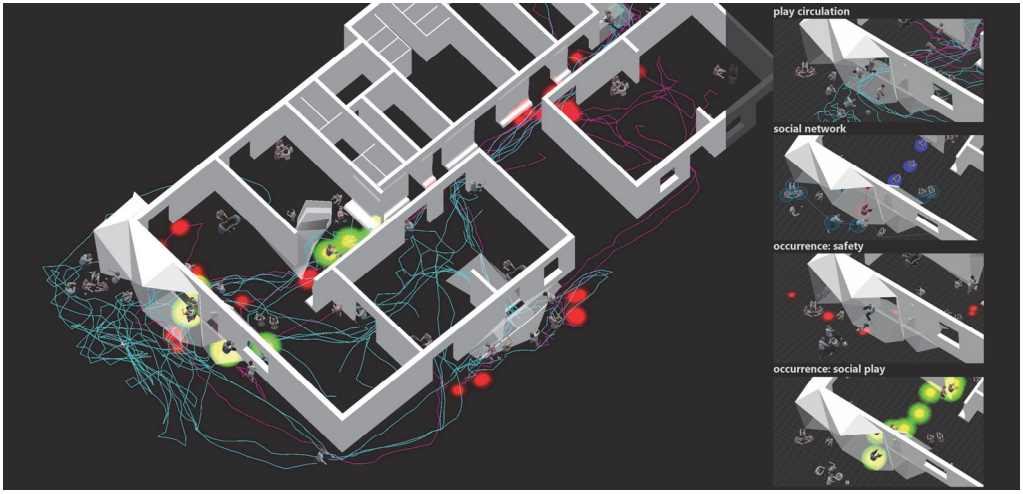


Figure 6
Analytics and empirical case analysis

Results

The design results of the experiment were analysed to quantify performance based on the previously mentioned criteria. RL agents were used to simulate (1) the frequency of physical and social play, (2) the duration of physical and social play and (3) safety (Table 1). We sampled five design cases (Figure 7).

Physical and social play: Most cases generally showed a significant increase in social play frequency, and a general increase for the duration of social play. In contrast, physical play frequency and duration did not display a consistent increase and varied depending on the design intentions. In general, the participants used visual cues to adjust the interaction and collision points and expanded the activity space to facilitate social play.

For example, Participant A changed the design goal from safety to social play by removing the skip floor and achieved a 71% increase in social play frequency. The duration of social play also increased by 5%. However, the significant increase in social play resulted in a 30% decrease in physical play frequency; the physical play duration remained unchanged.

Participant B increased physical play frequency by utilising the roof and removing steep climbable slopes, which resulted in a 67% increase in physical play duration and a 33% increase in social play frequency. However, after a significant increase in social play frequency, the duration of social play decreased by 10%.

Participant C's designs focused on physical and social play. To facilitate these, he simplified the overall shape of design for better children's interactions, resulting in an 87% increase in social play frequency. However, the significant increase in social play caused more interference during social play and resulted in a 20% decrease in the duration of social play. Also, removing skip floors led to a 28% reduction in physical play frequency.

Safety: The study found that most design results showed an increase compared to the initial alternative. Safety increased when the participants expanded the spatial capacity by removing inefficient and compact spaces.

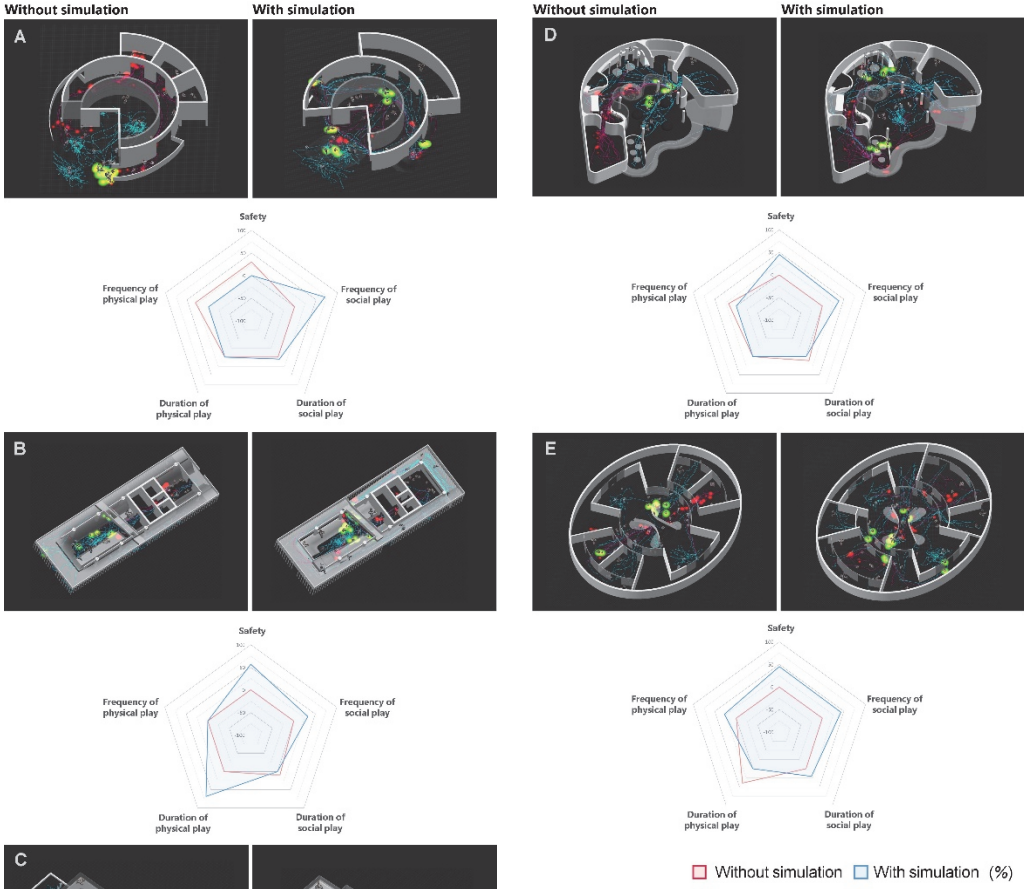
For example, Participant D coordinated the width of the entrances and the location of the opening with the children's movements to improve safety. As a result, safety increased by 45%. The frequency of social play also increased by 38%, while the number of physical plays and the duration of social play decreased by -18% and -11%, respectively.

Participant E coordinated the design based on the dimensions of the children's movements rather than on the standardised effective width. As a result, safety increased by 45%. The frequency of physical and social play also increased by 28% and 43%, respectively. The duration of social play increased by 22%, but the physical play duration decreased by -38%.

Participant	Without/With simulation	Frequency of physical play	Frequency of social play	Duration of physical play	Duration of social play	Safety
A	Without	100	7	0	-7	-7
	With	70	12	0	-7	-10
B	Without	84	9	-12	-10	-23
	With	84	12	-4	-11	-10
C	Without	87	8	-1	-5	-15
	With	63	15	-1	-6	-13
D	Without	110	8	0	-9	-11
	With	90	11	0	-10	-6
E	Without	69	7	-8	-9	-18
	With	88	10	-11	-7	-10

Table 1
Results of quantifiable learning performance

Figure 7.
Analysis of results



CONCLUSIONS AND DISCUSSIONS

In this study, we developed RL child agents and evaluated their effects in quantifying the kindergarten design performance. We confirmed that the RL child agents facilitated the holistic analysis and calculation of generative responses such as falling and serendipitous interactions.

Most design cases showed an increase in social play and safety by eliminating ineffective layouts and expanding the space available. We noted that unexpected agent behaviours helped discover hitherto unknown design affordances, thereby

leading to a better understanding of the social dynamics and child-oriented perspectives (Hong et al., 2016). Also, the results revealed that the participants targeted varied factors to enhance performance based on their design objectives. Through a holistic analysis and data-driven evidence, the agents assisted the participants in conducting design trade-offs by considering the intricate interplay between physical constraints and users' environmental responses for an optimal design (Kalay, 2004).

This study also addresses limitations and further development. The current RL child agent is based on a conceptual validation that abstracts children's play behaviour. Therefore, future research can conduct operative validation using authentic behaviour data in real-world settings. Furthermore, due to a small number of participants, a future study aims to ensure a valid sample size and employ quantitative statistical analysis approaches, such as the consensual assessment technique (CAT). Despite these limitations, we confirmed that this computational model can measure and predict the complex behaviours of children with quantifiable values, such as microinjuries and delays, responding to geometric properties. This study is expected to prove the empirical impacts of RL agents and to provide a systemic, evidence-based approach for child-oriented design.

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